

AKILAN MANIKANDAN

Backend & Automation Engineer | Python | FastAPI | RPA | Applied AI Systems

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SUMMARY

Backend and automation engineer with internship experience across 4 roles building production API services, RPA pipelines, and data ingestion systems that eliminated 8+ hours of daily manual effort. Shipped across Python, Java, and TypeScript using FastAPI, Spring Boot, NestJS, PostgreSQL, and MongoDB. Specializes in turning repetitive operational workflows into auditable, configuration-driven systems with hands-on work in applied ML, LangChain RAG pipelines, and campaign automation at scale.

TECHNICAL SKILLS

Languages	Python, Java
Backend	FastAPI, Spring Boot, Flask, Django REST Framework, NestJS, Next.js, React, GraphQL
Automation	Playwright, PyAutoGUI, Selenium, TagUI, Scrapy, n8n
Databases	PostgreSQL, MongoDB, Mongoose, SQLAlchemy, psycopg2
Data / ETL	Pandas, NumPy, Power BI REST API
Cloud / Storage	AWS, boto3
Auth / Security	JWT, Passport, bcrypt, AES, RSA, ABAC, OTP, MFA, TOTP
ML / AI	TensorFlow, Keras, Librosa, scikit-learn, LangChain, OpenAI API, Gemini API
Tools	Git, Maven, MS Office, Docker, Linux

INTERSHIPS

Software Engineer | *Clarivium Technologies*

Aug 2025 – Mar 2026

- Engineered **50+ Python automation scripts** converting recurring report-generation and file-handling processes into **reusable, configuration-driven workflows** eliminating an estimated **8+ hours of daily manual effort** across operations teams.
- Built **end-to-end RPA pipelines** using **Playwright** and **PyAutoGUI** covering login, **OTP/TOTP** prompts, date filtering, report export, and retry logic automating **12 recurring workflows** with **zero manual intervention** on success paths.
- Designed a **PostgreSQL-backed execution orchestration layer** with **idempotent state tracking** and audit logging reducing **duplicate pipeline runs by ~90%** and enabling full replayability of failed stages.
- Built and maintained **48** documented **n8n** workflows (21 branchwise KPI refresh + 25 dashboard freshness monitoring) integrating **Power BI** executeQueries API, **PostgreSQL** upserts, and notification steps replacing manual analytics validation for a 10-branch automotive client.
- Engineered **15+ Mautic-n8n automation workflows** on **AWS EC2** for a dealership campaign product, automating **WhatsApp and email outreach** via **AiSensy BSP APIs** across insurance renewals, service reminders, and test-drive follow-ups cutting campaign setup time **from hours to minutes**.

Backend Developer | *CleandeskAI*

Apr 2025 – Jul 2025

- Designed and shipped **5 Django REST Framework API endpoints** supporting document classification and workspace management in a production **PostgreSQL**-backed environment.
- Diagnosed and resolved **10+ backend defects** across API response handling and data model inconsistencies, improving endpoint reliability for active production users.

- Traced and corrected **PostgreSQL query logic** returning incorrect or missing data, restoring **data integrity for 20 users** without downtime.

Backend Engineer | *Simbiotik Technologies*

May 2024 – Jun 2024

- Built **15+ backend modules** for an **HRMS platform** using **NestJS, MongoDB, and Mongoose** implementing **CRUD APIs** across users, employees, tasks, and policy management with **role-guarded endpoints**.
- Secured **15+ API routes** by implementing **JWT- and Passport-based authentication** with **bcrypt password hashing, OTP-based MFA, and ABAC role guards** eliminating unauthorized access across all protected endpoints.

AI/ML Engineer | *Zidio Development*

Jul 2024 – Aug 2024

- Trained an **LSTM-based speech emotion recognition model** on the **TESS dataset**, achieving **97.6% validation accuracy** across **7 emotion classes** up from a **66% baseline** over 30 epochs through **MFCC feature engineering** and iterative hyperparameter tuning.

PROJECTS

Secure Data Storage & Controlled File Sharing System

Tech Stack: Java, Spring Boot, Flask, AES-256, RSA, ABAC, scikit-learn (Isolation Forest)

- Architected a **multi-role secure file-sharing system** in **Spring Boot** supporting upload, access-request, approval, and download workflows across **5 user roles**, implementing **AES-256 file encryption, RSA key exchange, and policy-based ABAC access control** with a full audit trail.
- Built a **Flask anomaly-detection microservice** using an **Isolation Forest model** and rule-based scoring to flag suspicious access behavior across **4 user roles** achieving **100% recall** and **0.611 F1 score** on test data, with real-time **email alerts** to administrators.

CityPulse — Event Recommendation Chatbot

Tech Stack: FastAPI, Next.js, React, MongoDB, LangChain, OpenAI GPT, Scrapy, Selenium

- Built a **FastAPI backend** and **Next.js frontend** with **Google OAuth sign-in** and a **LangChain RAG pipeline** over a **1,200+ event MongoDB corpus** delivering personalized recommendations with **conversation memory** and **<2 second average response time**.
- Developed a **Scrapy + Selenium scraping pipeline** covering **17 Chennai event categories**, normalizing **1,200+ event records** into **MongoDB** for real-time chatbot retrieval, with engagement tracking powering **popularity-based ranking**.

CERTIFICATIONS

Claude Code in Action	Anthropic	Mar 2026
AI Fluency for Students	Anthropic	Mar 2026
NLP for ML with Python: Advanced NLP Using spaCy & Scikit-learn	Skillsoft	Apr 2025
Python for Data Science	NPTEL	Apr 2025
Java Full Stack	Simbiotik Technologies	Jun 2024

EDUCATION

B.Tech — Computer Science and Engineering	2022 – 2026
SRM Institute of Science and Technology CGPA: 8.35/10	
Lalaji Memorial Omega International School	2020 – 2022
CLASS XII (CBSE) Percentage: 70%	
Hindu Senior Secondary School	2019 – 2020
CLASS X (CBSE) Percentage: 84%	